

Project Proposal

MedLedger: A Blockchain-Based Anti-Counterfeit Pharmaceutical Supply Chain with Relational and Graph Data Models

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1. Abstract

MedLedger is a proposed system that secures the pharmaceutical supply chain against counterfeit and substandard medicines by combining three complementary data technologies into one coherent architecture. A relational database (Oracle) serves as each organization's operational system of record; a Hyperledger Fabric permissioned blockchain provides a tamper-proof, multi-party record of drug custody; and a Neo4j graph database enables provenance tracing and counterfeit detection. Every unit of medicine carries a QR code encoding a unique package identifier, which acts as the shared key across all three layers. A consumer, pharmacist, or regulator can scan the code to instantly verify a pack's authenticity and view its complete custody history. This project deliberately demonstrates a *polyglot persistence* design: trust-critical, cross-organization data is stored on the blockchain, while detailed and private operational data remains in the database, allowing each technology to be used for what it does best. The project simultaneously satisfies the requirements of the Database Management Systems assignments and the Blockchain course.

2. Introduction and Background

Pharmaceutical products pass through a long chain of independent organizations — manufacturers, distributors, wholesalers, and pharmacies — before reaching a patient. In the current model, each organization maintains its own private database that it can modify at any time, and no single shared, trustworthy record of a product's journey exists. This fragmentation allows counterfeit products to be introduced into the chain and makes recalls slow, because reconciling records across multiple companies during an emergency can take days.

According to the World Health Organization, an estimated one in ten (approximately 10.5%) medical products circulating in low- and middle-income countries is either substandard or falsified, and countries spend an estimated US\$30.5 billion each year on such products. These medicines fail to treat disease and, in the case of antibiotics and antimalarials, contribute to avoidable deaths and to antimicrobial resistance. The problem is fundamentally one of *trust and traceability* across organizations that do not fully trust one another — precisely the problem that permissioned blockchain technology is designed to address. Industry has already begun adopting this approach: the MediLedger Network is

a live, industry-led permissioned blockchain used by major pharmaceutical manufacturers and distributors for supply-chain verification, demonstrating that the architecture proposed here is practical rather than merely theoretical.

3. Problem Statement

To design and implement a secure, verifiable pharmaceutical supply-chain system that (a) maintains a rich, normalized operational database of drugs, batches, quality tests, shipments, and dispensing; (b) records custody handoffs between organizations on an immutable, tamper-evident distributed ledger that no single party can falsify; (c) enables rapid tracing of any product's origin and rapid identification of every downstream recipient during a recall; and (d) allows end users to verify product authenticity through a simple scan.

4. Motivation

A conventional centralized database cannot solve this problem on its own, because the organizations involved are commercial competitors and regulators — a database owned by any one of them would not be trusted by the others, and its records could be altered after the fact. A permissioned blockchain removes the need for a trusted central party while guaranteeing that custody records cannot be tampered with retroactively. However, a blockchain alone is a poor tool for storing large volumes of detailed data, for protecting private information such as patient records, and for running complex analytical queries. Combining a blockchain with a relational database and a graph database therefore yields a system that is simultaneously trustworthy, efficient, privacy-respecting, and analytically powerful. This combination also makes the project an ideal single deliverable for both the Database Management Systems assignments and the Blockchain course.

5. Objectives

1. Design an Entity-Relationship (ER/EER) model for a pharmaceutical supply chain and normalize all relations up to Boyce–Codd Normal Form (BCNF).
2. Implement the schema in Oracle with PL/SQL business logic — triggers, stored procedures, and functions — to enforce expiry rules, validate custody transfers, and automate recalls.
3. Model the supply chain as a Neo4j property graph and demonstrate provenance-tracing and counterfeit-detection queries, with a comparative analysis justifying the graph model against a relational one for this use case.
4. Build a Hyperledger Fabric permissioned network with chaincode (smart contracts) that enforces immutable, multi-endorsed custody transfers between organizations.
5. Integrate the three layers through a shared package identifier and implement a QR-code-based authenticity-verification workflow.
6. Demonstrate a complete end-to-end scenario covering manufacture, distribution, dispensing, counterfeit detection, and recall.

6. Scope

In scope: the relational, blockchain, and graph layers; a QR-based verification application; synthetic sample datasets; and a demonstration scenario covering the full product lifecycle.

Out of scope: integration with physical IoT sensors; production-grade deployment and scaling; publication of a consumer mobile application; the use of real patient data (synthetic data will be used throughout); and payment or billing functionality.

7. Proposed System Architecture

The system is organized into three layers joined by a single shared identifier.

The **operational layer** is an Oracle relational database holding the complete detail of the supply chain: parties, drugs, batches, quality tests, packages, shipments, dispensing records, and recalls. It supports the complex, high-volume queries that day-to-day operations require.

The **trust layer** is a Hyperledger Fabric permissioned blockchain shared by four organizations — a manufacturer, a distributor, a pharmacy, and a regulator. It records only the custody events that must be trusted across organizational boundaries (batch created, shipped, received, dispensed, recalled). Each transfer must be endorsed by both the sending and the receiving organization, so no party can unilaterally fabricate a handoff, and once written, records cannot be altered.

The **analytical layer** is a Neo4j graph database that mirrors the ledger's custody events as a network of nodes and relationships, making it possible to trace a product's full journey and to detect fraud through pattern matching.

The unifying principle is that a blockchain ledger is itself a chain of transactions — a graph — so the graph database naturally mirrors the ledger, while the relational database holds the deep operational detail beneath both. The same `package_id`, encoded in each product's QR code, links a record across all three layers. Trust-critical, cross-organization data is placed on the blockchain; detailed and private data remains in the relational database.

8. Methodology and Module Breakdown

Module 1 — Database Design (maps to DA1). Produce the EER diagram, including a generalization of the three supply-chain parties into a common superclass; derive the relational schema; document the functional dependencies; and prove that each relation satisfies BCNF.

Module 2 — Relational Implementation (maps to DA2). Implement the schema in Oracle using SQL DDL, populate it with synthetic data, and develop PL/SQL logic: triggers to block expired or over-shipped stock and to cascade recalls, stored procedures to register batches and process recalls, and functions to verify authenticity and quantify recall impact.

Module 3 — Graph Provenance (maps to DA3). Model the supply chain in Neo4j, import a dataset, and implement Cypher queries that trace a batch's journey, list every pharmacy

downstream of a recalled batch, and flag counterfeits as impossible custody paths. Include the required comparative analysis of the graph model versus the relational model.

Module 4 — Blockchain (maps to the Blockchain course). Configure a Hyperledger Fabric network with four organizations, write chaincode implementing `createBatch`, `transferCustody`, `dispenseToPatient`, `issueRecall`, and `verifyPackage`, and enforce an endorsement policy requiring both sender and receiver approval for transfers.

Module 5 — Integration and Demonstration. Link the layers through the shared package identifier, implement the QR-verification workflow, and run the end-to-end demonstration scenario.

9. Technology Stack

Oracle 21c (or Oracle XE) with SQL Developer for the relational layer; Neo4j (Desktop or Aura) with Cypher for the graph layer; Hyperledger Fabric 2.x with chaincode written in Node.js for the blockchain layer; a QR-code library with a lightweight web or command-line front end for verification; and, optionally, a small Node.js or Python service that listens for ledger events and synchronizes them into Neo4j.

10. Expected Outcomes and Deliverables

The project will deliver an EER diagram with a documented BCNF normalization; a working Oracle database with PL/SQL triggers, procedures, and functions; a Neo4j graph with a set of provenance and fraud-detection queries and a comparative analysis; a running Hyperledger Fabric network with deployed chaincode and a QR-based verification tool; a demonstrable end-to-end counterfeit-detection and recall scenario; and a final report with a project presentation for each course.

11. References

1. World Health Organization. *Substandard and falsified medical products* (fact sheet) and *1 in 10 medical products in developing countries is substandard or falsified* (2017). who.int.
2. Ozawa, S. et al. *Prevalence and Estimated Economic Burden of Substandard and Falsified Medicines in Low- and Middle-Income Countries: A Systematic Review and Meta-analysis*. JAMA Network Open, 2018.
3. The MediLedger Network — pharmaceutical supply-chain blockchain and DSCSA verification. mediledger.com.
4. Hyperledger Fabric documentation. hyperledger-fabric.readthedocs.io.
5. Neo4j Cypher documentation. neo4j.com/docs.

(Replace bracketed placeholders with your details and format the references per your course's required citation style.)